

Anil

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### **Professional Summary:**

- 5+ years of experience in Data Engineering working with Python, PySpark, AWS, Azure, Hadoop, and Databricks.
- Strong background in building batch and streaming data pipelines using Spark, Glue, ADF, and Databricks.
- Hands-on experience with AWS services such as S3, Glue, EMR, Lambda, Step Functions, Redshift, and IAM.
- Good experience with Azure components including ADF, ADLS, Azure Databricks, Synapse, and Event Hubs.
- Worked extensively with Delta Lake, partitioning strategies, schema validation, and data lake architecture.
- Strong in SQL development, performance tuning, data modeling, and building ETL/ELT frameworks.
- Experience integrating data from databases, APIs, message queues, flat files, and on-prem systems.
- Good understanding of CI/CD, Git, DevOps, cloud security, and production monitoring.
- Comfortable working with Data Scientists, Analytics teams, and business teams to support data needs.

### **Technical skills:**

- **Programming & Scripting:** Python, PySpark, Scala, SQL, Shell Script
- **AWS:** Glue, EMR, Lambda, S3, Redshift, IAM
- **Azure:** ADF, ADLS, Event Hubs, Databricks, Synapse
- **Big Data Technologies:** Hadoop, HDFS, Hive, Spark SQL, Spark Streaming, Spark MLlib, Sqoop, Oozie, YARN
- **Data Processing & Analytics:** Apache Spark (Core, SQL, Streaming), PySpark, Databricks Notebooks, Delta Lake
- **Streaming & Messaging:** Apache Kafka, Kafka Connect, Azure Event Hubs, Spark Streaming
- **Cloud Platforms:** Microsoft Azure, AWS (EC2, S3, EMR), GCP (BigQuery)
- **Workflow Orchestration:** Apache Airflow, ADF Pipelines & Triggers, CRON
- **ETL / Data Integration Tools:** Azure Data Factory, Databricks ETL, StreamSets, IBM DataStage, Sqoop
- **Databases:** SQL Server, Oracle, MySQL, PostgreSQL, DB2, Teradata, MongoDB, HBase
- **Data Modeling & Warehousing:** Star Schema, Snowflake Schema, Dimensional Modeling, OLAP/OLTP
- **Visualization & Reporting:** Power BI, Tableau
- **Machine Learning & Analytics:** MLlib, Scikit-Learn, Pandas, NumPy (for ETL feature prep)
- **Version Control & Automation:** Git, GitHub, Azure DevOps, Jenkins (CI/CD)
- **Operating Systems:** Linux, Unix, Windows

### **Professional experience:**

**Client: Baidu, Mountain View, CA**

**Data Engineer**

**July 2024 - Present**

#### **Roles and responsibilities:**

- Built Glue ETL jobs in PySpark to process large batches of log files and partner data coming into S3; handled schema drift, corrupt files, and bad partitions.
- Designed S3 data lake layers (raw/processed/curated) and reorganized folder structures to fix slow Athena queries and inconsistent partition layouts.
- Developed Databricks notebooks for transformations, Delta Lake merges, incremental loads, and cleaning up messy semi-structured data.
- Tuned Glue and Spark jobs by adjusting shuffle partitions, enabling AQE, using broadcast joins, and fixing OOM errors during wide transformations.
- Created Step Functions and Glue Workflows to orchestrate multi-step ETL jobs and added CloudWatch alerts to track failures and lag.

- Used DMS for CDC ingestion from on-prem SQL Server to S3; resolved issues with late-arriving records and timestamp mismatches.
- Wrote Lambda functions in Python for automation tasks like file validation, metadata tagging, and triggering downstream jobs.
- Built Redshift loads for analytics teams and optimized queries by fixing distribution keys, sort keys, and vacuum operations.
- Implemented data quality checks such as row counts, type validation, null rule checks, and duplicate detection.
- Troubleshoot Glue job failures, Spark performance issues, missing partitions, and S3 permission errors across environments.

**Client: Pfizer, Memphis, TN**

**Dec 2022 – Jun2024**

**Data Engineer**

**Roles and responsibilities:**

- Built ADF pipelines to ingest SAP tables, LIMS files, QA batch records, and CSV extracts into ADLS Gen2.
- Developed PySpark jobs in Databricks to clean clinical lab results, manufacturing batch data, temperature logs, and other regulated datasets.
- Implemented strict data validation rules required for GxP/HIPAA (controlled values, reference lookups, audit fields).
- Converted legacy SSIS jobs into cloud-based ADF pipelines with dependency chains and checkpointing.
- Maintained Delta Lake tables for slow-changing clinical attributes, controlled schemas, and retention rules.
- Tuned Databricks clusters for batch-heavy workloads (partition tuning, caching dimension tables, reducing shuffles).
- Designed Synapse SQL models for QA/Regulatory reporting teams involving batch, lot, and material-level facts.
- Added lineage documentation, metadata tracking, and data quality scorecards for compliance audits.
- Set up Log Analytics dashboards to monitor pipeline failures, SAP file delivery issues, and schema mismatches.
- Worked closely with QA/Compliance teams to review dataset changes, validation checks, and business rules.

**Client: Qualcomm, Hyderabad India**

**Oct 2020- Aug 2022**

**Data Engineer**

**Roles and responsibilities:**

- Built streaming pipelines using Apache Kafka and Spark Streaming to process telecom/device telemetry data.
- Developed ETL pipelines using PySpark, Spark SQL, and Airflow for batch ingestion and transformation.
- Managed datasets in HDFS, Hive, and HBase for storage, query processing, and analytics use cases.
- Used Kafka Connect and Sqoop to ingest data from Oracle, SQL Server, and PostgreSQL into Hadoop.
- Created PySpark UDFs for parsing device logs, performing aggregations, and applying complex business rules.
- Processed unstructured logs and stored processed data in MongoDB/HBase for analytics teams.
- Integrated MLlib and scikit-learn models into Spark pipelines for device performance insights.
- Processed large volumes of semi-structured/unstructured telemetry data using PySpark and Kafka for real-time insights.
- Performed cluster performance tuning for Spark jobs by optimizing batch intervals, parallelism, and memory usage.
- Used AWS S3 and on-prem Hadoop for hybrid storage and long-term archival.
- Implemented data validation scripts to ensure accuracy of processed telecom datasets.

**Educations Details:**

**Masters in Advance Data Analytics (University of North Texas 2024).**

**Bachelors in Business Administration (Math's and Statistics Major) from Vishwa Vishwani Business School 2020.**